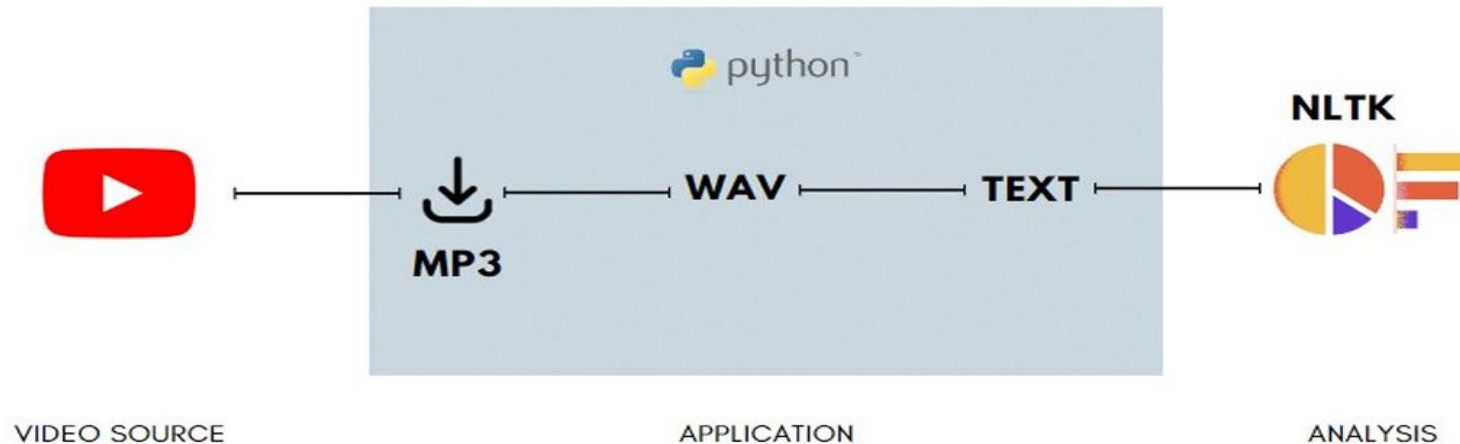


Tried Data Analysis on a YouTube video.

#Python application to automate the analysis process.

1. Extract the audio from a YouTube video and download as MP3 file to local disk.
2. Convert the MP3 file to WAV file.
3. Convert the wav file to the text using the Google Recognition engine.
4. Using NLTK analytics of text.



```
[3]: # Deep analysis of Dr. S Jaishankar speech At UNSC video source --> Yotube
```

```
[*]: #To download the youtube video as mp3 and convert to wav
```



```
from pytube import YouTube
import os

yt = YouTube(str(input("Enter URL of youtube video: \n ")))
video = yt.streams.filter(only_audio=True).first()
destination = r"C:\Users\arpit\Downloads\New folder"
out_file = video.download(output_path=destination)
base, ext = os.path.splitext(out_file)
new_file = "podcast" + 'wav'
os.rename(out_file, new_file)
print(yt.title + " TEMPhas been successfully downloaded.")
```

Enter URL of youtube video:

```
[13]: import speech_recognition as sr
from pydub import AudioSegment
r = sr.Recognizer()

with sr.AudioFile(r"C:\Users\arpit\anaconda3\Youtube Analysis\clean.wav") as source:
    audio_text = r.record(source, duration=400)

try:
    text = r.recognize_google(audio_text)
    print(text)

except Exception as e:
    print("Error:", e)
```

I know if the flowchart is Excellency Subramanyam Jaishankar minister of external affairs of India president excellencies ladies and gentleman namaskar from the people of Bharat decades since the founding of this unique party not just to prevent the dignity of every human being to stand as good neighbours United Kingdom of plastic free download history since the founding of the Year colonization advanced the world began to return to its natural diversity the membership of the United Nations for the organisations role and rabbit Cruise significantly in the era of globalisation father development goals to centre stage even as climate change emerged as a shared priority trade acquired greater prominence food and health access were recognised as essential to global World Bank security operations and promoted the government discussion became a natural platforms today how has the yuva lived up to Expectations and just look at the state of the world significant conflicts on the way one in Ukraine and the other in the Middle East station don't even make the news the slow progress of the SDG agenda 2030 presence sorry picture on climate change the reality is of recycled commitments and creative accounting is climate action itself is question what hope is that for climate justice open discrimination in access to vaccines to travel energy and food security have been the first casualties of conflict and disruption especially since 2022 better of societies insulated themselves by having the first call the resource trust once scramble to survive only to hear sanctimonious lectures that after non market practices game rules and regions the resulting concentration expose the world to leverage on top of that volatility and uncertain market access result is a growing competition weather from limited sources of supply or over dependence on a particular market economic conditions have other dimensions of height and Technology control is one the group and supply chains and Critical min

erals is another shipping of connectivity is no less sensitive protection of TC length has become a challenge restricting the evolution of a global workplace is an issue makes a complete International cooperation not less headed in the direction as members of the UN we meet that is not a formulation of the inherent diversity of our understandable we have different history tradition it is imperative faster understanding and respect political interference and economic pressures when narratives of freedom is truly appreciate madam president excellent we must also firmly face up to threats countering terrorism is a particular priority because its synthesis

[15]: `import nltk`

`nltk.download('punkt')`

```
[nltk_data] Downloading package punkt to  
[nltk_data] C:\Users\arpit\AppData\Roaming\nltk_data...  
[nltk_data] Unzipping tokenizers\punkt.zip.
```

[15]: True

[17]: `import nltk`

`print(len(text), "Words long speech by Dr. S. Jaishankar (Minister of External Affairs)")`

2659 Words long speech by Dr. S. Jaishankar (Minister of External Affairs)

[19]: `import nltk`

`nltk.download('punkt_tab')`

```
[nltk_data] Downloading package punkt_tab to  
[nltk_data] C:\Users\arpit\AppData\Roaming\nltk_data...  
[nltk_data] Unzipping tokenizers\punkt_tab.zip.
```

[19]: True

```
[20]: #tokenization
      from nltk.tokenize import word_tokenize
      words=word_tokenize(text)
      words
```

```
[20]: ['I',
      'know',
      'if',
      'the',
      'flowchart',
      'is',
      'Excellency',
      'Subramanyam',
      'Jaishankar',
      'minister',
      'of',
      'external',
      'affairs',
      'of',
      'India',
      'president',
      'excellencies',
      'ladies',
      'and',
      'gentleman',
      'namaskar',
      'from',
      'the',
      'people',
      'of',
      'Bharat',
      'decades',
      'since',
      'the',
      'founding',
```

```
'terrorism',  
'is',  
'a',  
'particular',  
'priority',  
'because',  
'its',  
'synthesis']
```

```
[22]: #frequency of words  
from nltk.probability import FreqDist  
fdist=FreqDist()  
for word in words:  
    fdist[word.lower()] +=1  
fdist  
list(fdist)
```

```
[22]: ['the',  
      'of',  
      'and',  
      'is',  
      'to',  
      'a',  
      'as',  
      'in',  
      'world',  
      'climate',  
      'on',  
      'that',  
      'since',  
      'not',  
      'access',  
      'have',  
      'market',  
      'we',  
      'president',
```

```
counting ,  
'terrorism',  
'because',  
'synthesis']
```

```
[23]:  
# how many time he uses word "attack" and "terrorist"  
print(fdist["threats"],"&",fdist["terrorism"],"Times Respectively ")  
  
1 & 1 Times Respectively
```

```
[24]:  
# how many different word Dr. S jaishakar used in his speech  
len(fdist)
```

```
[24]: 262
```

```
[25]: # most common word he used  
print(fdist.most_common(10))  
  
[('the', 32), ('of', 24), ('and', 16), ('is', 14), ('to', 13), ('a', 9), ('as', 5), ('in', 5),  
('world', 4), ('climate', 4)]
```

```
[26]: nltk.download('stopwords')
```

```
[nltk_data] Downloading package stopwords to  
[nltk_data] C:\Users\arpit\AppData\Roaming\nltk_data...  
[nltk_data] Unzipping corpora\stopwords.zip.
```

```
[26]: True
```

[27]:

```
# Stop words : have no meaning in analytics  
from nltk.corpus import stopwords  
st=stopwords.words('english')  
st
```

```
was ,  
'wasn',  
"wasn't",  
'we',  
"we'd",  
"we'll",  
"we're",  
'were',  
'weren',  
"weren't",  
"we've",  
'what',  
'when',  
'where',  
'which',  
'while',  
'who',  
'whom',  
'why',
```


[29]:

```
# remove the stop words from speech and look about what the actully speech was
all_words=list(fdist)
means=[]
for i in all_words:
    if i in st:
        pass
    else:
        means.append(i)
means
```

```
'market',
'president',
'founding',
'united',
'history',
'natural',
'diversity',
'even',
'change',
'priority',
'food',
'global',
'security',
'one',
'first',
'supply',
'particular',
'economic',
'loss'
```

[31]:

```
# conclusion : Dr. S Jaishankar talked about :  
for i in range(10):  
    if len(means[i])>3:  
        print(means[i])
```

world
climate
since
access
market
president
founding
united
history
natural

[]: